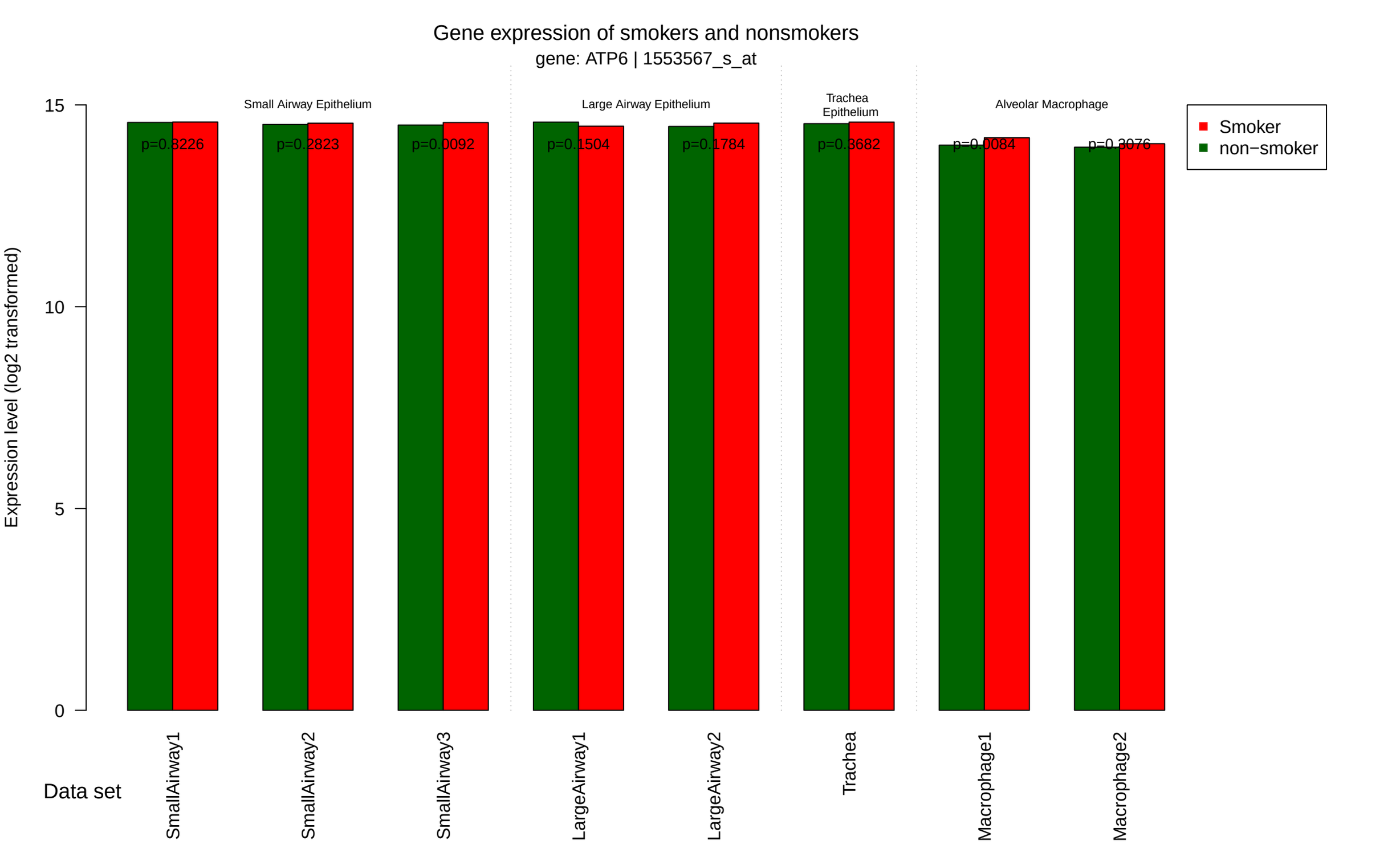
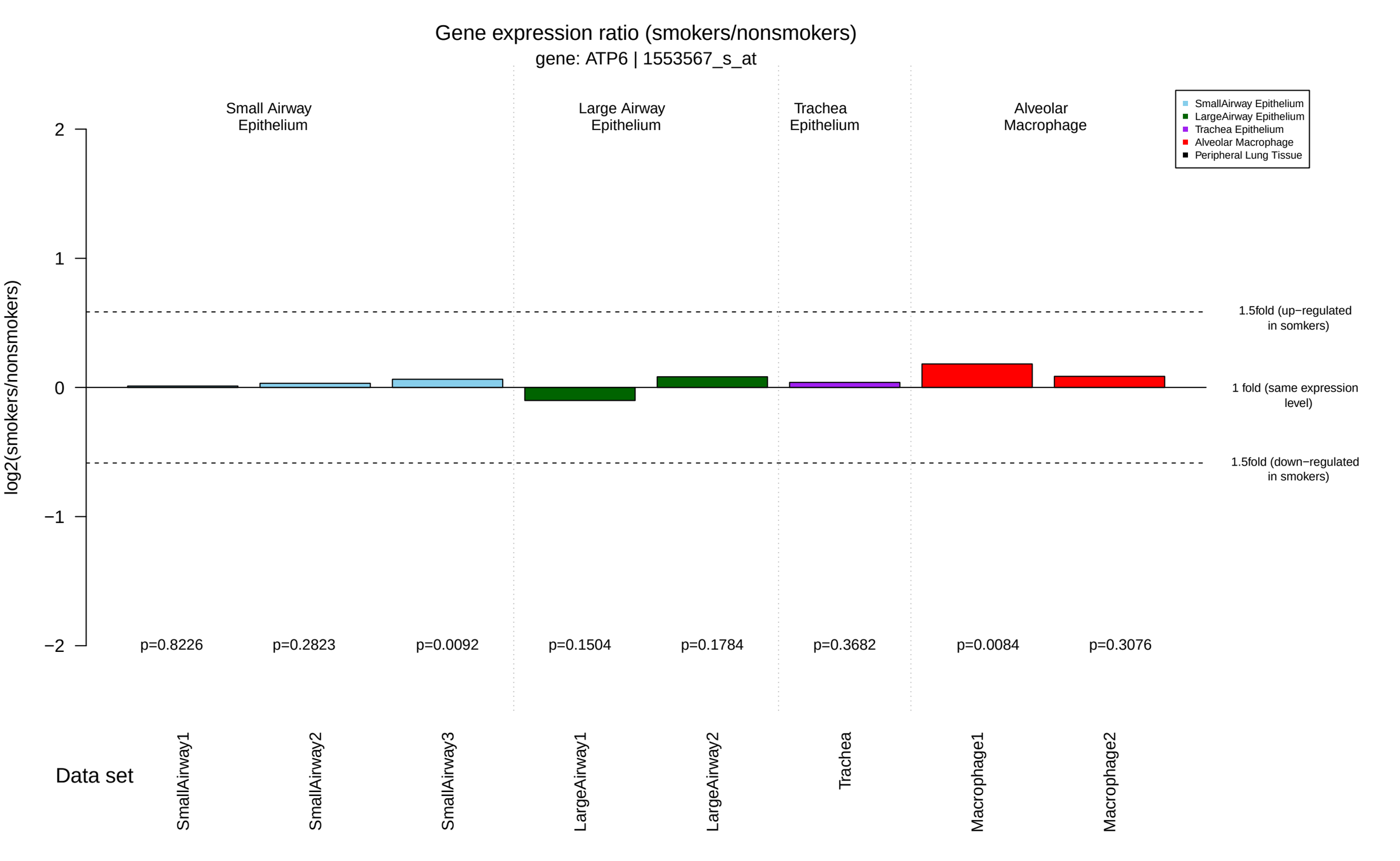
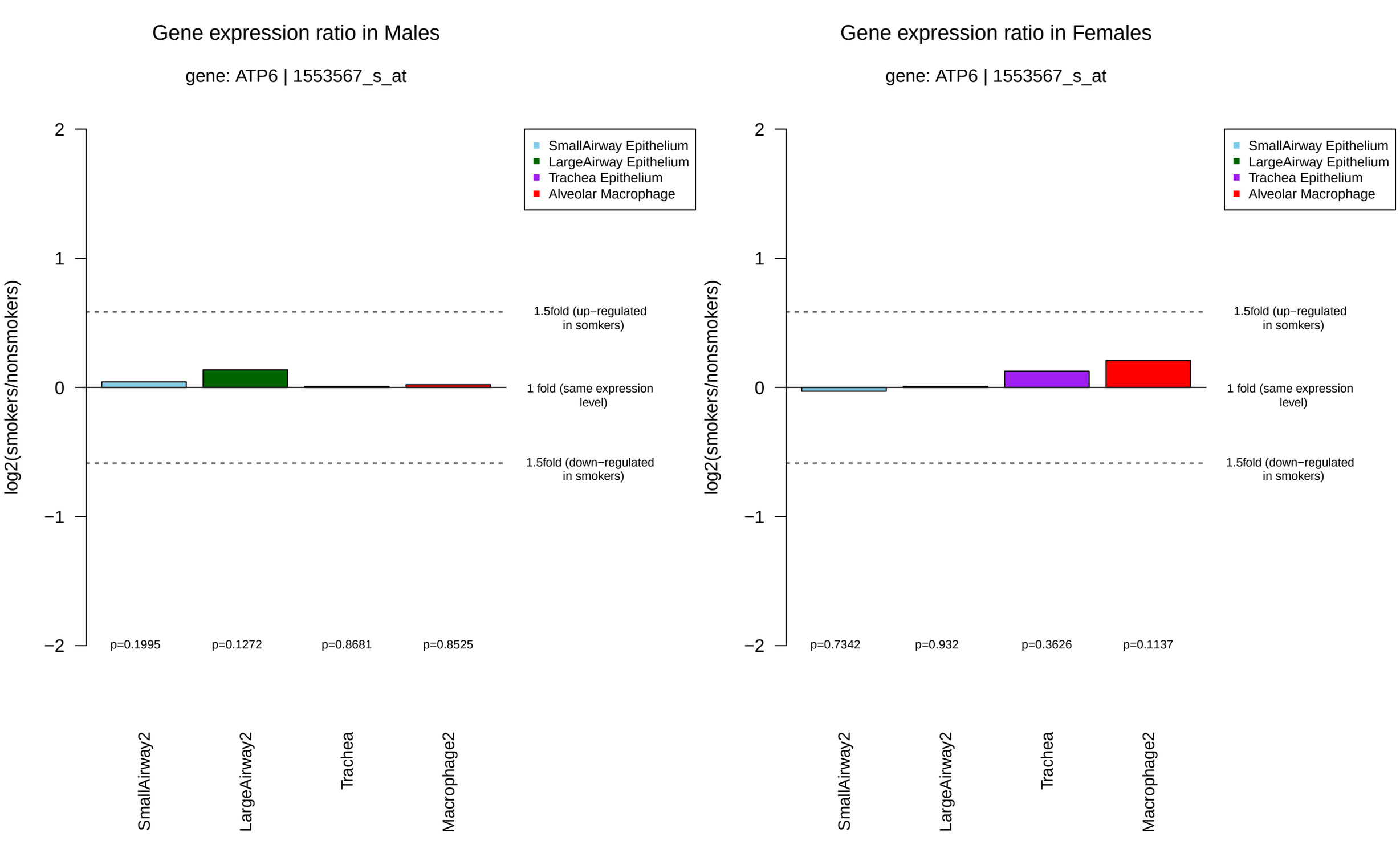
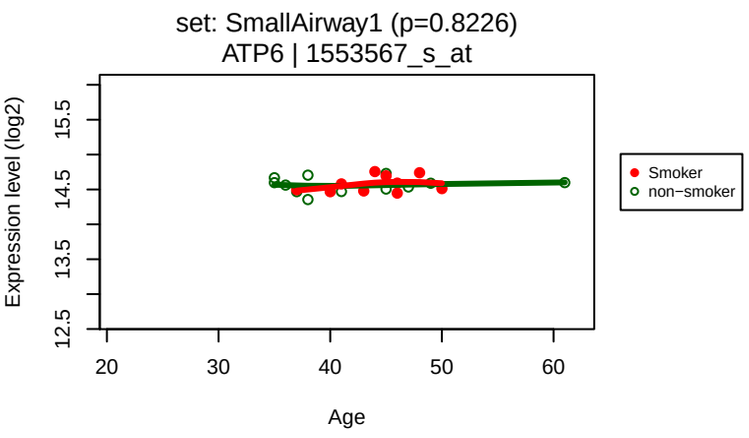
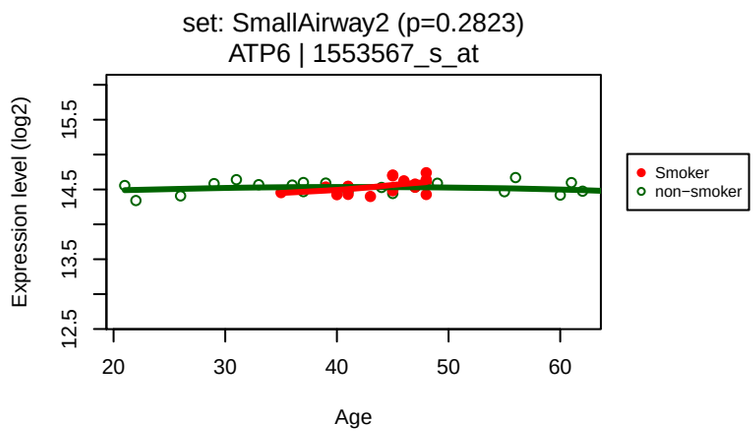
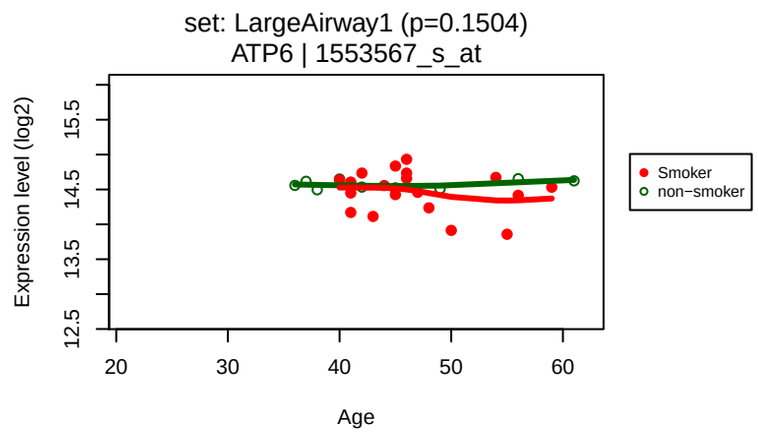

Gene expression of smokers and nonsmokers
gene: ATP6 | 1553567_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8226
p=0.2823
p=0.0092
p=0.1504
p=0.1784
p=0.3682
p=0.0084
p=0.3076
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

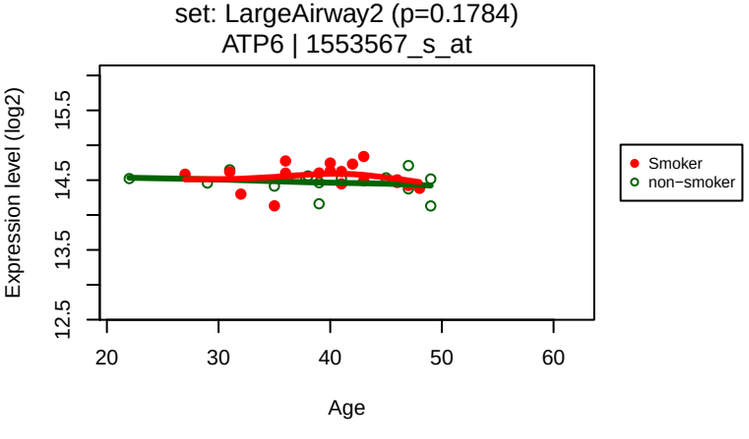

Gene expression ratio (smokers/nonsmokers)
gene: ATP6 | 1553567_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8226
p=0.2823
p=0.0092
p=0.1504
p=0.1784
p=0.3682
p=0.0084
p=0.3076
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

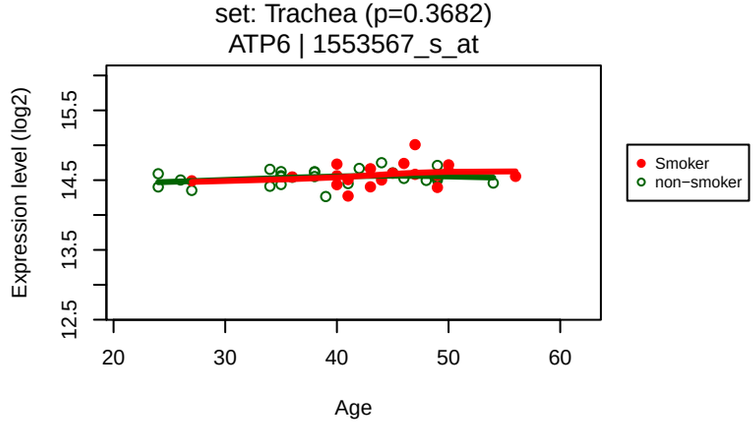

Gene expression ratio in Males
gene: ATP6 | 1553567_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1995
p=0.1272
p=0.8681
p=0.8525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP6 | 1553567_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7342
p=0.932
p=0.3626
p=0.1137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

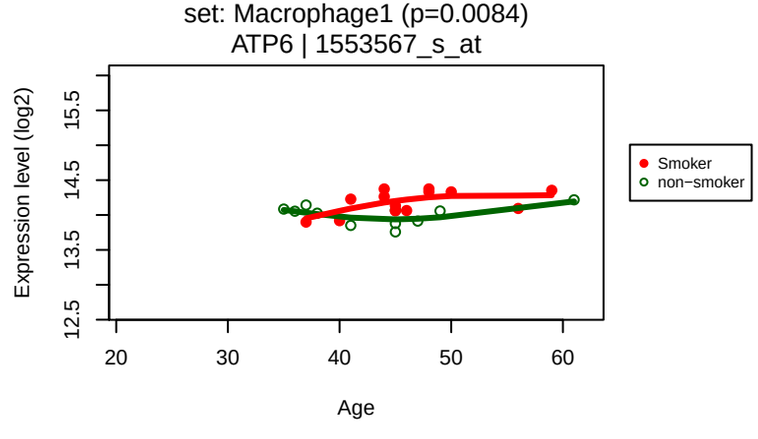

set: SmallAirway1 (p=0.8226)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

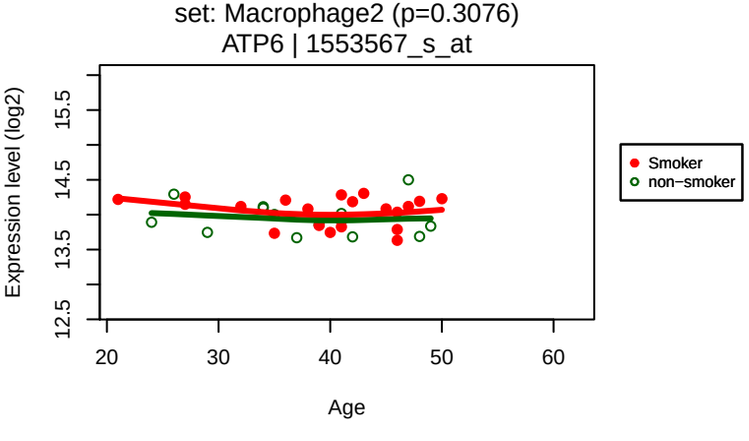

set: SmallAirway2 (p=0.2823)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

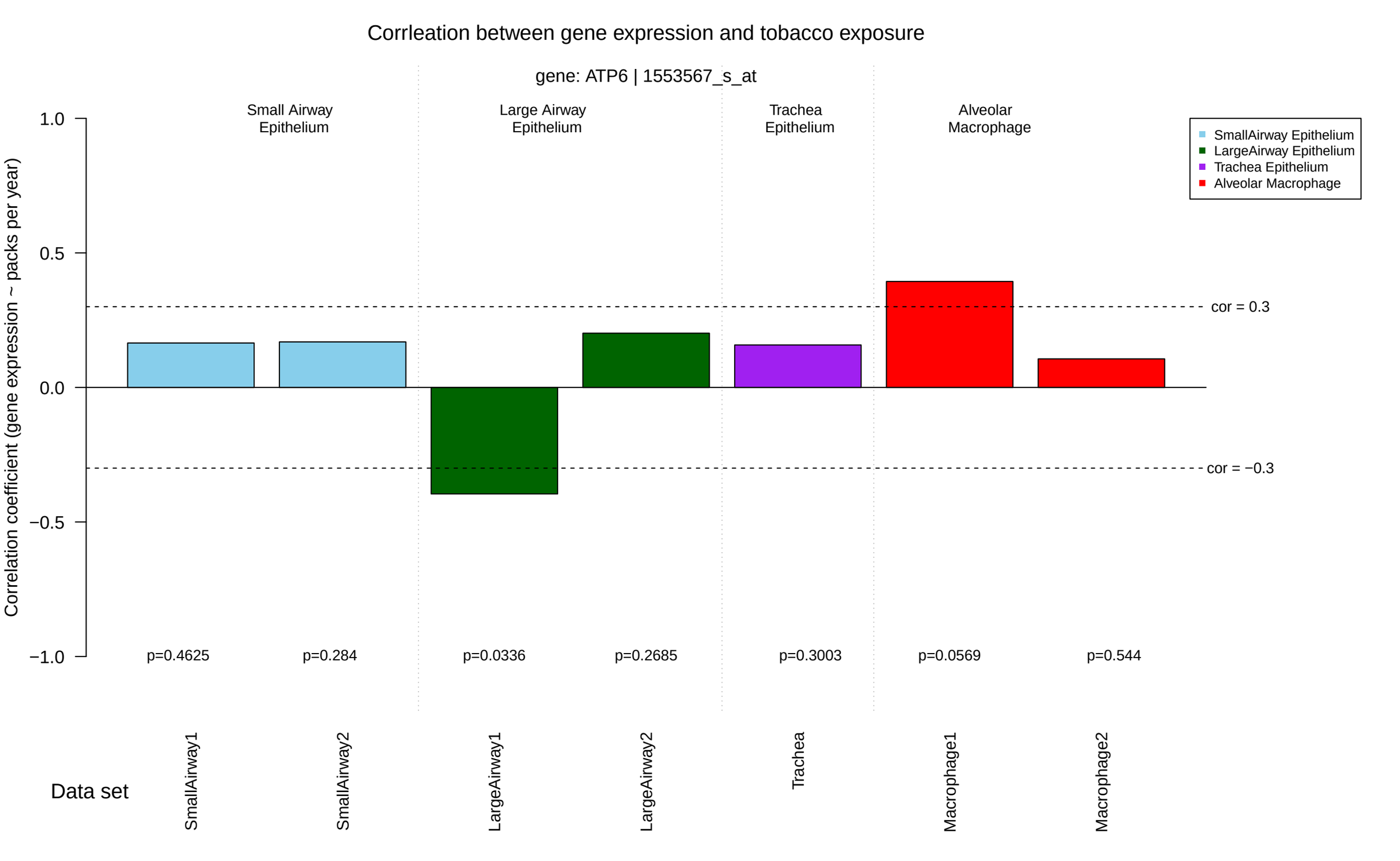

set: LargeAirway1 (p=0.1504)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1784)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3682)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0084)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3076)
ATP6 | 1553567_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP6 | 1553567_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4625
p=0.284
p=0.0336
p=0.2685
p=0.3003
p=0.0569
p=0.544
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2